Pranav Varshney

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Education

University of Michigan — Aug 2026

B.S. Computer Science — Ann Arbor, MI

Courses: Data Structures, Machine Learning, Advanced Algorithms, Operating Systems, Distributed Systems

Experience

Google — May 2026 - Aug 2026

Software Engineering Intern | Cost Optimization — San Francisco, CA

- *Incoming*: Implementing a context-aware AI cost optimizer for GCP workloads using GNN dependency mapping

Uber — Jan 2026 - Apr 2026

Software Engineering Intern | Delivery Matching — Sunnyvale, CA

- Enhanced uRoute observability for BYOC + uETA predictions, plumbing data across primary + fallback paths
- Launched cache sync/async kafka logging for Uber Eats predictions, handling 100k+ requests/s at ∼5ms latency
- *Current*: Refactoring deepETA models to a eyeball vs unified checkout + postorder stage publishing evaluations

Scale AI — July 2025 - Oct 2025

Generative AI Intern | Evaluation Pipelines — Remote

- Developing retrieve + rerank RAG pipeline to automate LLM inputs for data annotation via candidate filtering
- Reduced annotation error by using a novel multi-stage reranking (LLM-as-a-judge + Adjucation) pipeline

Amazon Web Services — May 2025 - July 2025

Software Engineering Intern | Inference Platform — Bellevue, WA

- Created a terminal service for mapping 1M+ Bedrock endpoints to adjacent AWS Services for troubleshooting
- Developed a real-time system health monitoring framework, yielding an on-call response time reduction of 70%
- Built containerization tooling for u systems through AWS Copilot, enabling support over 10,000 queries/wk

Research

Precision Imaging × AI Laboratory | Prof. Joyce Wang — Oct 2025 - Dec 2025

- Engineered models for automated segmentation of cardiac chambers + quantification of ejection fraction
- Built a standardize CVD diagnosis platform, targeting interoperability with portable ultrasound devices

Self Operating Networks Laboratory | Prof. Victor Liu — Jan 2025 - May 2025

- Designed AST-aware transformer using constrained attention mechanisms, achieving 92% code generation accuracy
- Implemented contrastive learning pipeline for automated dataset expansion, increasing training samples by 250%

Projects

Cvrve

- *Discontinued*: helped develop the infrastructure observability behind job board (prev. cvrve) used by 200k+ users

Sharded Key-Value Store | Go, Paxos, Concurrency

- Built a fault-tolerant, sharded key-value store using Paxos for consensus + coordination between replica groups

Skills

Languages: Python, C++, Go, Java, C#, Typescript, Javascript, SQL

Technologies: Docker, Pandas, NumPy, Plotly, PyTorch, Kafka, Spark, Flask, Next, React, Node, Tailwind CSS

Concepts: Inference Engineering, Parallel Programming, Big Data, Natural Language Processing, Computer Vision, Mechanistic Interpretability, Cloud Computing, Quantum Computing, Computer Security, Full-Stack Development